Gene Expression Changes in Cancer Cells Under Drug Treatment

Alex Katz & Dvir Drori | Computational Bioskills | 2026

1. Introduction

Cancer drugs often fail to kill all tumor cells. Some cells survive by entering a dormant state called persistence, and can later resume growth. This project is based on Oren et al. (Nature, 2021), who treated PC9 lung cancer cells with the EGFR inhibitor Osimertinib and tracked gene expression using single-cell RNA sequencing. By Day 14, surviving cells split into distinct subpopulations with different proliferation states and gene expression programs.

2. The Data

The scRNA-seq data covers four timepoints: Day 0 (before treatment), Day 3, Day 7, and Day 14. At Day 14, cells were sorted by FACS into cycling, moderate-cycling, and non-cycling groups. Each CSV file contains one sample where rows are cells (barcodes), columns are genes, and values are UMI counts (RNA molecules per gene per cell). The dataset has 11 files totaling ~56,000 cells. We focused on 15 genes from five categories: cell cycle, stress/survival, housekeeping, apoptosis, and drug resistance.

3. Exploratory Figure

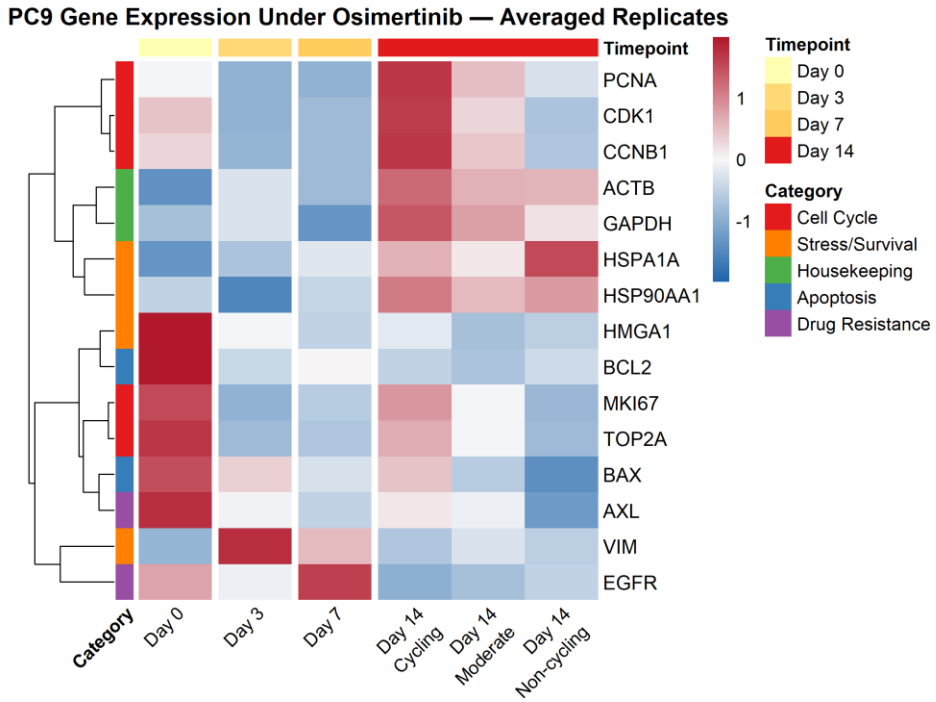


Figure 1: Heatmap of mean expression of 15 genes across 6 conditions (Day 0 through Day 14), ordered chronologically. Colors are row-wise z-scores: red = above gene mean, blue = below. Annotations show gene category and timepoint.

This heatmap gives a broad view of how expression changes over time. Cell cycle genes crash almost to zero by Day 3, showing the drug acts within days. VIM (a mesenchymal marker) peaks at Day 3, suggesting an early transient stress response. From Day 3 to Day 7 there is little change, but by Day 14 the cells diverge: cycling cells reactivate proliferation genes while non-cycling persisters show elevated stress (HSPA1A). Housekeeping genes (ACTB, GAPDH) remain stable throughout, validating the comparison. We chose a heatmap because it compares many genes across many conditions at once, and z-score scaling allows comparison between genes with different absolute expression levels.

4. Explanatory Figure

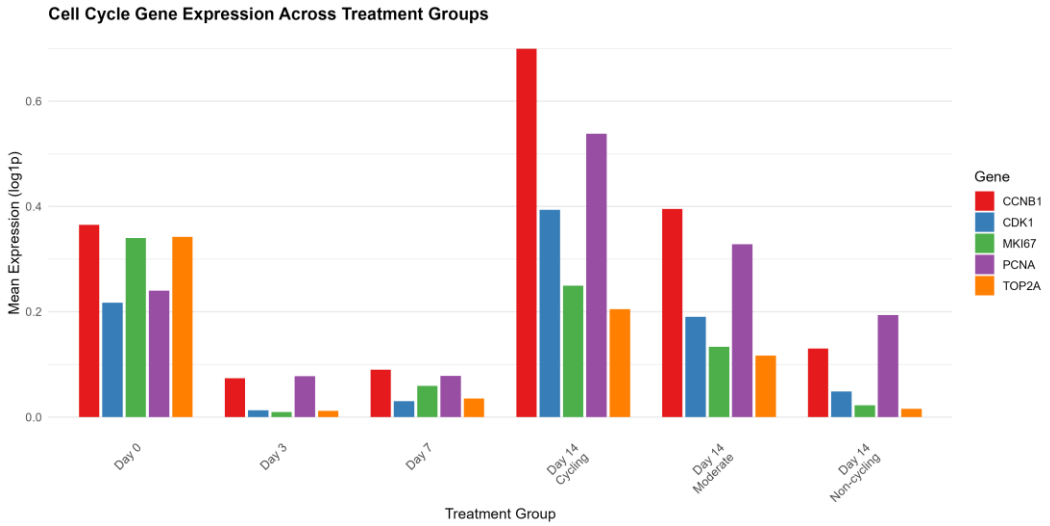


Figure 2: Grouped bar plot of five cell cycle genes (CCNB1, CDK1, MKI67, PCNA, TOP2A) across six conditions from Day 0 to Day 14. Replicates are averaged.

We asked: do cell cycle genes progressively decrease from cycling to non-cycling cells? The plot confirms this and reveals the full timeline. Day 3 is the lowest point - the drug suppresses proliferation almost completely within three days. Day 7 shows a slight recovery, suggesting early adaptation. At Day 14, cycling cells exceed Day 0 levels for CCNB1 and PCNA, meaning a resistant subpopulation emerged. From cycling to moderate to non-cycling, all genes decline stepwise, showing the arrest is a gradient. We chose a bar plot because it makes the quantitative decline across groups clear for each gene individually.

References

Oren, Y., Tsabar, M., Cuoco, M.S. et al. (2021). Cycling cancer persister cells arise from lineages with distinct programs. Nature, 596, 576-582.

Gutfreund, A. et al. (In preparation). Global dysregulation and drug persistence in bacteria.